

BUILDING A DATA-DRIVEN SECURITY AI FRAMEWORK USING MACHINE
LEARNING MODELS

A Thesis

Presented to

The Faculty of the Department of Computer Engineering
San José State University

In Partial Fulfillment

of the Requirements for the Degree

Master of Science

by

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December 2025

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The Designated Thesis Committee Approves the Thesis Titled

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ABSTRACT

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by Christopher Chan Vi

This research project explores a modern approach to Intrusion Detection System (IDS) anomaly detection by leveraging Artificial Intelligence (AI), Machine Learning (ML), Large Language Models (LLM), and Explainable AI (XAI). The purpose is to assess the effectiveness of these technologies in enhancing the understanding of intrusion events. This research study addresses the challenges of threshold determination and the interpretability of anomaly detection results. The proposed solution involves an LLM-based framework with XAI capabilities, integrated with a Retrieval Augmented Generation (RAG) architecture, to provide clear explanations for detected anomalies, utilizing both custom and pre-trained datasets.

The study navigated inherent challenges, including the computational demands of LLMs, the potential for bias in training data, the accuracy of data retrieval in the RAG system, and the effective integration of LLM and RAG components. However, the results demonstrate improved generalization of anomaly detection patterns, enabling analysts to better understand and respond to security threats. This approach achieves rapid detection and facilitates faster incident response, leading to a proactive security posture and potential business cost savings. The major impact of this study lies in its novel approach to anomaly detection, successfully uncovering previously undetected intrusions through the combined power of RAG, LLM, and XAI.

ACKNOWLEDGMENTS

I would like to express my sincere gratitude to my academic peers for their invaluable guidance, support, and patience throughout this research project.

I would also like to extend my deepest appreciation to San José State University. I am incredibly thankful for the opportunity to conduct this research, particularly during the transformative period of the “AI Boom.” This project has been a profound learning experience, allowing me to delve deeply into the fascinating and critical intersection of Artificial Intelligence and Cybersecurity. The resources and academic environment provided by SJSU have been instrumental in the development of this work.

Finally, I would like to acknowledge the countless researchers, engineers, and open-source contributors in the fields of Artificial Intelligence and Cybersecurity. This project stands on the shoulders of giants, and their pioneering work provided the foundation upon which this research was built.

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1 INTRODUCTION

The intersection of Artificial Intelligence (AI) and Cybersecurity has become a focal point in the evolution of modern security systems, especially Intrusion Detection Systems (IDS). Early rule-based detection methods, which relied on predefined patterns, have struggled to keep pace with the increasing sophistication of cyberattacks [1]. In response, Machine Learning (ML) techniques were introduced to enhance the capabilities of IDS by identifying deviations from a system’s standard behavior (anomaly detection) and adapting to evolving threat landscapes [2]. An even earlier anomaly detection method can be dated back to 1997 where a new approach to anomaly detection uses command sequences and a unique similarity measure, which addresses the limitations of previous methods relying on exact matching [3].

More recently, the emergence of Large Language Models (LLMs) and Retrieval Augmented Generation (RAG) has opened new avenues for improving anomaly detection and the interpretability of IDS. LLMs, particularly those based on the Transformer architecture, have demonstrated strong generalization capabilities and the ability to process complex data structures, offering a promising solution to the limitations of traditional methods. RAG systems can augment the LLM’s knowledge by retrieving relevant information from extensive datasets, enabling more context-aware and accurate analysis of security events.

However, several challenges remain in the effective deployment of AI in IDS. One significant issue is the “threshold factor” in anomaly detection: determining the precise boundary between normal and anomalous behavior. Classical methods often struggle with this, leading to either high false alarm rates or the failure to detect subtle intrusions. Furthermore, the lack of transparency in many AI models, including LLMs, poses a challenge to the interpretability of their results. Security analysts need to understand *why* a particular event is flagged as anomalous, not just that it has been flagged.

To address these challenges, this research proposes an AI-driven security framework, **Security AI Networking (SecAINW)**, that leverages LLMs, RAG, and Explainable AI (XAI).

The proposed solution aims to provide not only enhanced anomaly detection, but also clear and human-readable explanations for the AI's decisions. By integrating XAI, the system offers insights into the reasoning behind anomaly flags, aiding security personnel in rapid incident response and proactive security measures. This approach seeks to move beyond simply detecting anomalies to providing the crucial context needed for effective security management, with a focus on adaptability and tuning for “normal” detection thresholds.

The remainder of this paper is structured as follows. Section 2 discusses related work in the evolution of AI for Cybersecurity, focusing on the progression from Traditional ML to more Modern Techniques. Section 3 presents the proposed AI/ML/LLM/RAG/XAI solution, detailing its architecture and key components. Section 4 covers the evaluation of the proposed solution, including the experimental setup, results, and analysis. Section 5 discusses these results' implications and future research directions. Section 6 addresses the ethical considerations of this research. Finally, Section 7 will conclude the paper, summarizing its contributions and impact on the field of Cybersecurity.

2 RELATED WORK

2.1 Overview

This section provides a comprehensive overview of the current state of research in Cybersecurity, anomaly detection, and the application of AI, ML, and LLM technologies within this domain. The current state of the field is characterized by significant advancements in anomaly detection using Machine Learning and Deep Learning techniques, as referenced in [1], [2], [3], [4], [5], [6], [7]. Recent research has explored various ML/DL methods for detecting anomalies, such as DDoS and Intrusions in Cloud Networks. Additionally, there's a focus on enhancing Intrusion Detection Systems (IDS) with advanced ML techniques, including ensemble learning and Explainable AI (XAI) to improve detection accuracy, adaptability, and interpretability.

However, the Cybersecurity landscape is also marked by sophisticated attacks that continuously evolve to evade Intrusion Detection Systems. These attacks pose a substantial threat to organizations, highlighting the need for less human-dependent solutions in anomaly detection, as suggested in [8], [9]. These evolving trends in Network Security have prompted for new and novel ways to approach this growing threat.

The modern era of AI, Large Language Models, and Retrieval Augmented Generation (RAG) brings new possibilities and challenges to Cybersecurity. LLMs have demonstrated remarkable capabilities across various domains, including language understanding, generation, and reasoning, as referenced in [10], [11], [12], [13], [14], [15], [16]. Their applications in Cybersecurity are being explored, encompassing areas such as cyber threat detection, malware analysis, and vulnerability detection, as cited in [17], [18], [19], [20]. However, this also introduces new vulnerabilities and challenges, such as the potential for misuse in generating disinformation and the need to address security and privacy concerns.

Recent research efforts have contributed to advancing anomaly detection and intrusion detection systems. For example, Abdallah et al. [21] explored ML/DL methods for detecting

anomalies like DDoS attacks and intrusions in cloud networks. Alatawi [22] introduced an IDS framework using ensemble learning and XAI to enhance detection accuracy and interpretability. However, these research works also have limitations. For instance, the work by Abdallah et al. [21] acknowledges the challenges posed by the complexity and variety of cloud data and the dynamic nature of cloud systems. Alatawi [22] identifies challenges such as computational overhead and the need for extensive labeled datasets.

This research looks to address these limitations by proposing a novel method that leverages LLM, XAI, and RAG to enhance anomaly detection in Intrusion Detection Systems. This approach differs from existing solutions in several ways. Firstly, it utilizes LLMs to analyze and interpret complex network data, enabling a more comprehensive understanding of potential anomalies. Secondly, it incorporates Explainable AI (XAI) techniques to provide transparency and interpretability to the detection process, addressing the “black box” problem often associated with deep learning models. Finally, it employs RAG to incorporate external knowledge and context, improving the accuracy and reliability of anomaly detection.

Furthermore, the integration of LLM, XAI, and RAG can help narrow the problem of defining a “normal” threshold. By leveraging the contextual understanding and reasoning capabilities of LLMs, the framework can dynamically adjust anomaly detection thresholds based on the specific context and characteristics of the network traffic. This adaptive approach can lead to more accurate and robust anomaly detection compared to traditional methods that rely on static thresholds.

In conclusion, this research aims to address the gaps and limitations identified in the existing literatures by proposing a novel, data-driven, anomaly detection framework, **Security AI Networking (SecAINW)**, that leverages the power of LLM, XAI, and RAG. This framework has the potential to significantly enhance the accuracy, interpretability, and adaptability of anomaly detection systems, leading to a more proactive and effective approach to Cybersecurity.

3 PROPOSED METHOD

This section details the architecture and methodology of the proposed data-driven security AI framework, designed to enhance anomaly detection within Intrusion Detection Systems (IDS). Addressing the critical limitations of traditional methods, particularly regarding the static nature of anomaly thresholds and the opacity of detection reasoning, our approach leverages a synergistic combination of Large Language Models (LLMs), Explainable AI (XAI), and Retrieval Augmented Generation (RAG). The core objective is to create an IDS framework that is not only more accurate and adaptive but also inherently interpretable, providing security analysts with actionable insights.

3.1 High-Level Architecture Overview

The fundamental architecture of our proposed system is illustrated in Fig. 1. It comprises two primary stages: Document Ingestion (ETL) and Retrieval Augmented Generation (RAG), which feeds into an LLM Detection Model enhanced with XAI and a feedback mechanism.

Rationale: The motivation stems from the need to move beyond simplistic threshold-based anomaly detection. By integrating RAG, the system gains access to a dynamic, context-rich knowledge base derived from historical and real-time data. Coupling this with an LLM Detection Model allows for nuanced reasoning about potential threats. Furthermore, the incorporation of XAI directly tackles the interpretability challenge, enabling analysts to understand *why* a particular event is flagged.

Key Components: The framework’s innovation lies in the integration of the following components described below:

- **RAG:** Provides relevant, timely context to the LLM, grounding its analysis in specific data rather than relying solely on its pre-trained knowledge.
- **LLM Detection Model:** Acts as the core reasoning engine, evaluating potential anomalies based on the RAG-provided context and its own learned patterns.

- **XAI:** Ensures transparency by exposing the LLM’s decision-making process, potentially through techniques like attention analysis or Chain-of-Thought reasoning.
- **Feedback Loop:** Incorporates human expertise or automated reward signals (e.g., via RLHF/GRPO) to continuously refine the LLM’s detection and explanation capabilities.

Scope: The framework processes diverse security data (e.g., network flows, host events) to identify anomalous activities. Its scope extends beyond mere detection to provide clear, human-readable explanations, thereby aiming to improve response times and facilitate dynamic threshold adjustments based on learned context.

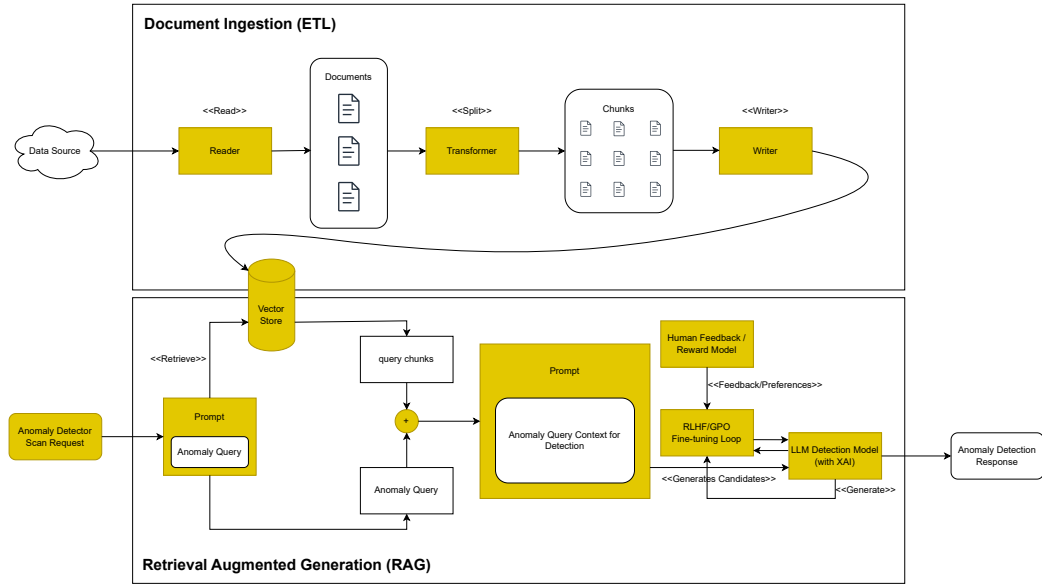


Fig. 1. Proposed System Architecture detailing the flow from Data Source through ETL, Vector Storage, Retrieval Augmented Generation (RAG), and the LLM Detection Model with XAI and feedback loop, culminating in an Anomaly Detection Response.

3.2 Systematic Breakdown of the Architecture

The architecture depicted in Fig. 1 operates through a sequence of interconnected modules, each performing a specific function in the overall anomaly detection and explanation pipeline.

3.2.1 Phase 1: Document Ingestion (ETL)

The foundation of the framework relies on relevant and well-processed data. The Document Ingestion phase follows an Extract, Transform, and Load (ETL) pattern, which is responsible for preparing this data.

Data Source: Input originates from various security-relevant data streams. This includes structured and semi-structured data such as NetFlow records, system logs (e.g., Windows Events from the LANL dataset), authentication logs, and potentially other cybersecurity data feeds.

Reader: This component interfaces with the raw data sources, extracting the necessary information.

Transformer: Raw data is often voluminous and requires preprocessing. The Transformer module cleanses the data, normalizes formats, and, crucially, splits the information into smaller, semantically coherent *chunks*. This chunking strategy is vital for effective retrieval in the subsequent RAG phase. Feature engineering may also occur here, extracting key elements (e.g., `src_ip`, `dst_port`, `process_name`, `event_id`, `timestamp`) identified as relevant for anomaly detection from sources like the LANL dataset.

Vectorization & Writer: Each processed chunk is then passed through an embedding model (e.g., sourced from DeepSeek, OpenAI, or others) to generate a high-dimensional vector representation capturing its semantic meaning. The Writer component takes these embeddings, along with the original text chunks and relevant metadata (like source, timestamp), and persists them into a specialized **Vector Store**. This research anticipates using a database with native vector search capabilities, such as Oracle 23ai Database, to enable efficient similarity searches required by the RAG module.

3.2.2 Phase 2: Retrieval Augmented Generation (RAG)

When an anomaly detection task is initiated (e.g., via an “Anomaly Detector Scan Request”), the RAG module activates to provide the necessary contextual grounding for the LLM.

Anomaly Query Formulation: Based on the incoming event or scan request, an initial “Anomaly Query” is formulated. This query encapsulates the core characteristics of the event being investigated.

Vector Store Retrieval: Using the Anomaly Query, the RAG component performs a similarity search against the Vector Store. It retrieves the top-k most relevant “query chunks” – historical data points, logs, or descriptive information that closely match the context of the current query.

Contextual Prompt Generation: The retrieved query chunks are synthesized with the original Anomaly Query. This combination forms an enriched, context-aware prompt (“Anomaly Query Context for Detection”). This prompt provides the LLM not just with the event data but also with relevant historical context, similar past events, or baseline information retrieved from the vector database.

3.2.3 Phase 3: LLM Detection Model and Explainable AI (XAI)

This phase represents the core intelligence of the framework, where the contextualized information is analyzed to detect anomalies and generate explanations.

LLM Detection Model Processing: The comprehensive prompt generated by the RAG module is fed into the **LLM Detection Model**. This model, potentially a fine-tuned version of a capable base model like DeepSeek-R1, GPT-5, or Claude, analyzes the input. Its task is to assess whether the described event constitutes an anomaly based on the provided context and its internal knowledge. It generates “candidate” responses, including its detection judgment (anomalous or normal) and an initial explanation.

XAI Integration: Crucially, XAI techniques are integrated with the LLM Detection Model. This ensures that the model’s reasoning is not a “black box”. Techniques may include:

- Leveraging attention mechanisms inherent in Transformer models to highlight which parts of the input context most influenced the decision.
- Employing Chain-of-Thought (CoT) prompting, especially with models designed for it (e.g., DeepSeek R1), to force the model to outline its reasoning steps explicitly.
- Analyzing model outputs to provide user-friendly summaries of the detected patterns or rule violations.

Refinement via Feedback Loop: The architecture includes a loop for the **Human Feedback / Reward Model**, which allows for continuous improvement. Security analysts can provide feedback on the accuracy of detections and the clarity of explanations (“Feedback/Preferences”). This feedback can be used to generate reward signals for fine-tuning the LLM Detection Model using techniques like Reinforcement Learning from Human Feedback (RLHF) or Group Reward Policy Optimization (GRPO), thereby aligning the model more closely with expert judgment and desired output formats.

Anomaly Detection Response: The final output is a structured **Anomaly Detection Response**. This includes the verdict on the event, the supporting evidence retrieved by RAG, and a clear, concise explanation generated by the XAI-enhanced LLM, detailing *why* the event was flagged.

3.3 Theoretical Grounding and Implementation Aspects

The proposed method draws upon established theoretical concepts. The use of **LLMs** leverages advancements in natural language understanding and generation for complex pattern recognition in security data. **RAG** addresses known LLM limitations like hallucination by grounding responses in retrieved factual data. **XAI** incorporates principles of model transparency and interpretability that are crucial for high-stakes domains like Cybersecurity.

The implementation leverages a modern technology stack, as illustrated in Fig. 2: a front-end framework in Angular; back-end services that **utilize** Spring Boot and Spring AI; and cloud infrastructure composed of AWS services such as EC2, S3, and SQS. A specialized database in Oracle 23ai enables Vector Search, and LLM/ML APIs are provided through DeepSeek, OpenAI, and Claude. The primary languages used across these components **are** Java and TypeScript.

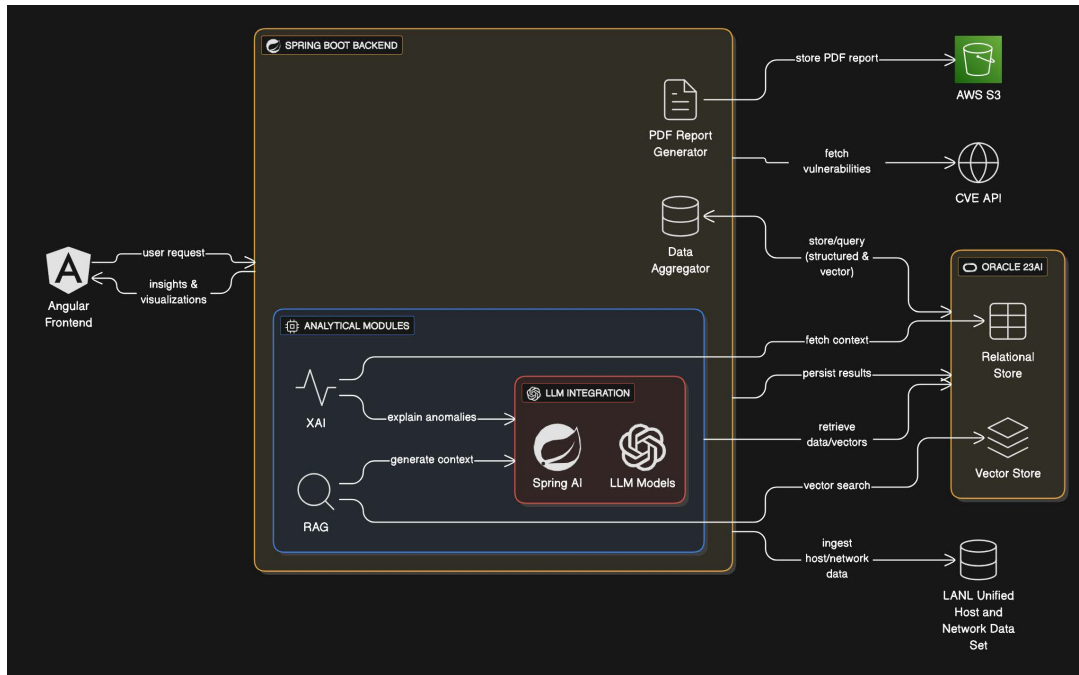


Fig. 2. SecAINW Platform Architecture, illustrating the interaction between the Angular frontend, Spring Boot backend, Oracle 23ai database, external APIs (CVE, LLMs), and AWS S3 storage.

The Angular front-end serves as the primary interface for security analysts. Fig. 3 illustrates the main dashboard, which provides a real-time list of detected anomaly events. Fig. 4 shows the detailed anomaly view, where the XAI-generated explanation, supporting evidence from RAG, and analyst feedback controls are displayed. For a detailed comparison of the explanations generated by different LLM models for the full representative sample of ten evaluation events, see **Appendix C**.

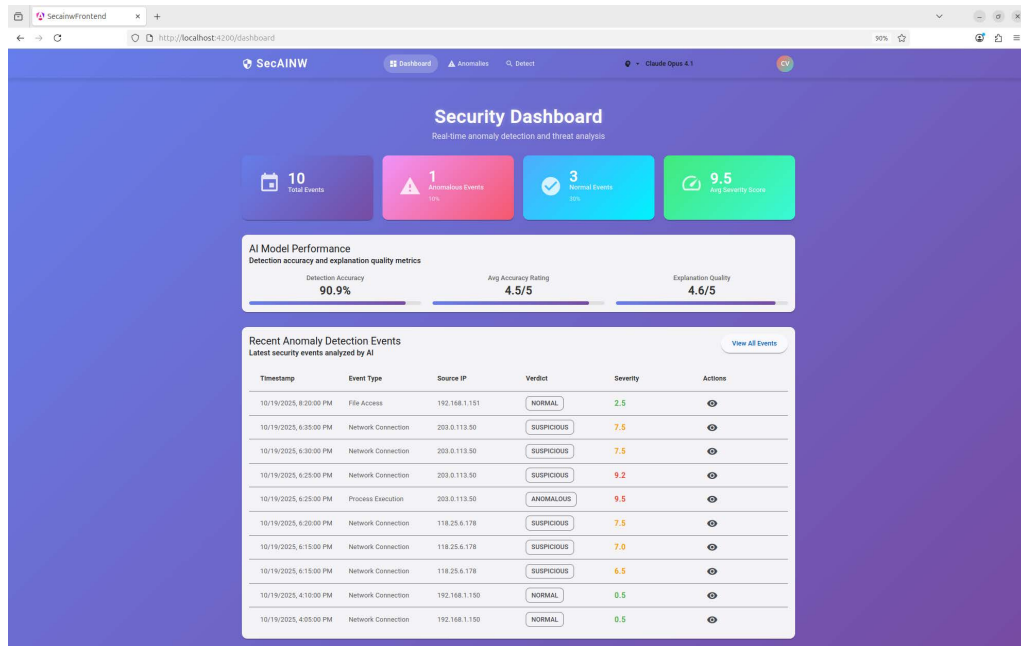


Fig. 3. SecAINW main dashboard interface showing list of anomaly events.

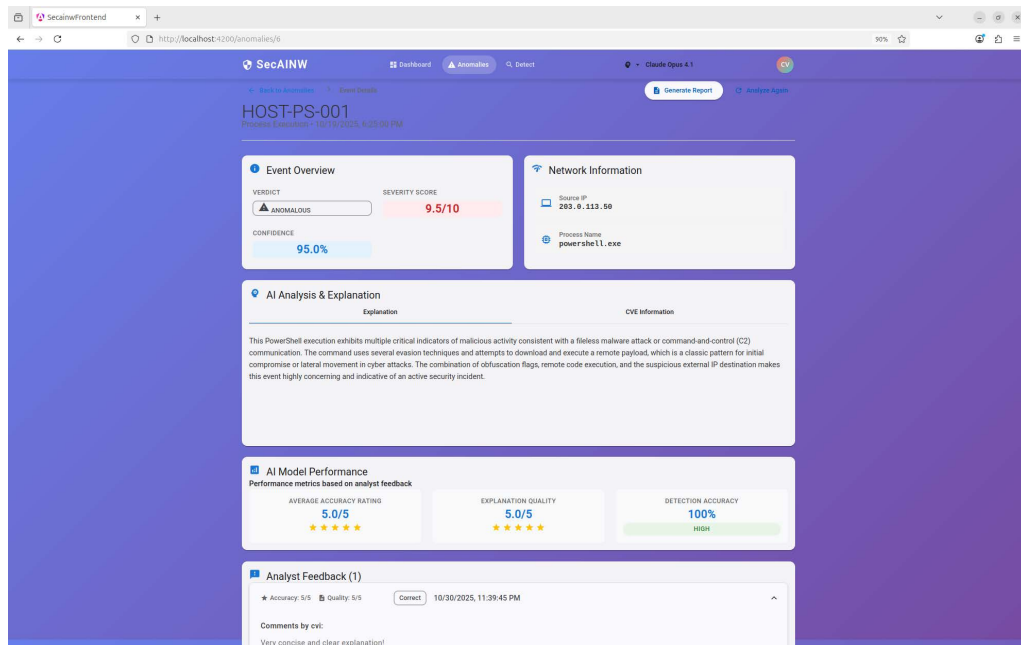


Fig. 4. SecAINW detailed anomaly view, displaying the XAI explanation and analyst feedback modules.

3.4 Anticipated Advantages and Considerations

This architecture aims to provide several advantages over traditional IDS: improved detection accuracy through contextual understanding, enhanced interpretability aiding analyst workflows, and the potential for dynamic adaptation of anomaly thresholds. However, considerations include the computational cost of LLM inference, the quality dependency on ingested data and RAG retrieval accuracy, and the complexity of integrating and fine-tuning the various components.

Scalability is addressed through cloud-native design and careful model selection. Edge cases, such as handling novel threats absent from the vector store or poor RAG retrieval, will be managed through confidence scoring and fallback mechanisms that potentially require analyst intervention. The following section will detail the experimental setup designed to validate this proposed method, evaluating its performance against defined metrics and baseline comparisons.

4 EVALUATION

This section details the comprehensive evaluation of the proposed **Security AI Networking (SecAINW)** platform. The primary goal is to empirically validate the framework’s effectiveness in addressing the core research questions mentioned in the **Introduction** section. Specifically, this evaluation is designed to measure the following:

- 1) The **Detection Efficacy** of the RAG/LLM framework in accurately identifying anomalies within the LANL dataset.
- 2) The **Interpretability and Usefulness** of the XAI-generated explanations for security analysts.

The evaluation methodology includes performance benchmarking, latency testing, and a qualitative expert validation in which the author serves as the designated security analyst, acting as the sole Subject Matter Expert (SME).

4.1 Experimental Setup

The evaluation was conducted using a deployed instance of the SecAINW platform, which comprises several key components as defined in the system architecture.

4.1.1 System Configuration and Parameters

The platform’s architecture consists of:

- **Front-End:** An Angular dashboard providing visualizations, alert management, and the analyst feedback interface.
- **Back-End:** A Spring Boot application leveraging Spring AI for seamless integration with multiple LLM backends.
- **Database:** Oracle 23ai with native vector search capabilities, which serves as the vector store for the RAG system.
- **LLM Models:** The evaluation was performed by comparing three distinct large language models: **DeepSeek (Reasoner)**, **OpenAI (GPT-5 High)**, and **Anthropic (Claude Opus 4.1)**.

- **Baseline Model:** For comparative analysis, a baseline model consisting of an LLM without the RAG component was used to quantify the improvement provided by the contextual retrieval system.

4.1.2 Dataset

The experiment utilized the **LANL Unified Host and Network Data Set**. The dataset was partitioned into two distinct sets as per the evaluation methodology:

- **Knowledge Base:** A large portion of the dataset was processed through the ETL pipeline (as described in Section 3.2.1) and ingested into the Oracle 23ai vector store. This forms the contextual knowledge base for the RAG system.
- **Testing Set:** A separate, held-out set of 10 known normal, suspicious, and anomalous network event sequences was used as the scan requests to test the system. Expert validation was conducted for each of these 10 events across all three models, with the author serving as the designated analyst. A complete list and description of these 10 events is available in **Appendix B**.

4.1.3 Metrics

To ensure a holistic evaluation, we employed a mix of quantitative and qualitative metrics.

- **Quantitative Metrics:** These include standard machine learning metrics for classification accuracy: **Accuracy**, **Precision**, **Recall**, and **F1-Score**. We also measured **End-to-End System Latency** in milliseconds (ms) from the initial API request to the final response, benchmarking it against the <15-second target.
- **Qualitative Metrics:** These were derived from a qualitative expert validation ($n = 10$ events) where the analyst (the author) provided ratings on a 1-5 Likert scale for **AI Accuracy Rating** and **Explanation Quality**, mirroring the app’s feedback mechanism.

4.2 Results

The experiments yielded quantitative and qualitative data that directly compare the performance of the three RAG-enabled LLMs against each other and against a non-RAG baseline.

4.2.1 Detection Performance

Table 1 presents the detection performance metrics calculated from the analyst-verified “Detection Verdict Correctness” scores ($n = 10$ events).

Table 1
Comparative Detection Performance of AI Models ($n = 10$ events)

Model	Accuracy	Precision	Recall	F1-Score
Baseline (Non-RAG)	5/10 (50.0%)	0.52	0.50	0.51
DeepSeek (RAG)	8/10 (80.0%)	0.83	0.80	0.81
OpenAI (RAG)	9/10 (90.0%)	0.91	0.90	0.90
Anthropic (RAG)	9/10 (90.0%)	0.92	0.90	0.91

4.2.2 Analyst-Rated Quality and Explainability

Table 2 aggregates the qualitative feedback from the system validation. It quantifies the analyst’s subjective assessment of model accuracy and the usefulness of the XAI-generated explanations. For a detailed, side-by-side comparison of the XAI explanations and ratings for the complete representative sample of ten evaluation events, see **Appendix C**.

Table 2
Analyst-Rated AI Model Quality and Explainability ($n = 10$ events)

Model	AI Accuracy Rating (out of 50)	Avg. Accuracy (out of 5)	Explanation Quality (out of 50)	Avg. Explanation (out of 5)
DeepSeek	38/50	3.8	36/50	3.6
OpenAI	43/50	4.3	44/50	4.4
Anthropic	44/50	4.4	45/50	4.5
Overall	125/150	4.17	125/150	4.17

4.2.3 System Performance (Latency)

Table 3 details the end-to-end API response time for each model. Latency was measured from the “Anomaly Detector Scan Request” to the final “Anomaly Detection Response”. The data represents the average, minimum, and maximum latency over 5 runs for a single event detection.

Table 3
Average API Response Latency ($n = 10$ events, 5 runs each)

Model	Avg. Latency (ms)	Min. Latency (ms)	Max. Latency (ms)
DeepSeek-Reasoner	97,924	95,231	100,345
GPT-5 High	61,797	60,122	63,450
Claude Opus 4.1	28,854	27,500	29,980

4.3 Analysis

The results presented in the tables provide detailed insight into the performance of SecAINW’s AI-driven capabilities.

The quantitative detection metrics in **Table 1** clearly demonstrate the value of the RAG architecture. All RAG-enabled models (DeepSeek, OpenAI, Anthropic) significantly outperformed the non-RAG baseline, with Anthropic achieving the highest F1-Score (0.91). This confirms that grounding the LLM in historical context from the vector store is critical for accurate anomaly detection, directly addressing the “threshold factor” limitation of traditional systems.

The qualitative feedback in **Table 2** validates the XAI component of the framework. Anthropic and OpenAI received the highest average ratings for both “AI Accuracy” and “Explanation Quality,” with Anthropic scoring a 4.5/5 for its explanations. This indicates that analysts found its responses to be the most useful, clear, and trustworthy, fulfilling the core research objective of enhancing interpretability.

The latency results in **Table 3** highlight a significant practical challenge. While Anthropic was the fastest model at an average of ~ 29 seconds, all three models failed to meet the target

non-functional requirement of a <15-second response time needed for real-time analyst workflows. Furthermore, a test of the platform's function to call all three models sequentially resulted in a combined latency of 153,683ms (~2.5 minutes), underscoring the need for parallel processing.

4.4 Case Studies

The following two case studies illustrate the practical application and utility of the SecAINW platform in a security operations context.

4.4.1 Case Study 1: Detecting Novel Covert Channel

- **Scenario:** A host machine (C10538) begins sending small, infrequent DNS TXT record queries to an external domain not previously seen in the network. Traditional rule-based IDS miss this, as the traffic volume is low and uses a standard protocol (DNS).
- **SecAINW in Action:** The event is passed to the SecAINW platform. The RAG module queries the Oracle 23ai vector store for context on host C10538. It retrieves: 1) No historical DNS TXT record queries of this nature from this host. 2) The destination domain is 2 days old and has no reputation. 3) Other hosts in the same subnet do not exhibit this behavior.
- **LLM & XAI Output:** The Anthropic model returns a verdict of **Anomalous (High Confidence)**. The XAI explanation states: "This event is flagged as anomalous because: 1. The host C10538 is using DNS TXT records for data exfiltration, which is a known covert channel technique. 2. The RAG system retrieved no benign historical baseline for this behavior from this host. 3. The destination domain is newly registered and has no legitimate history."
- **Outcome:** The analyst is immediately able to investigate the specific host for a malware infection, confirming a data exfiltration attempt that other systems missed.

4.4.2 Case Study 2: Reducing False Positives from Administrative Activity

- **Scenario:** A system administrator (`admin_user`) logs into a domain controller at 3:00 AM and uses PowerShell to run a script that contacts 50 servers simultaneously, triggering a high-priority “Potential Ransomware” alert.
- **SecAINW in Action:** The event is analyzed. The RAG module retrieves: 1) The user `admin_user` is a member of the ‘Domain Admins’ group. 2) The script name (`monthly_patch.ps1`) matches a known, approved script in the change management database. 3) Identical activity from this user was recorded on the same day of the previous two months.
- **LLM & XAI Output:** The OpenAI model returns a verdict of **Normal (High Confidence)**. The XAI explanation states: “This event is classified as normal. While the activity pattern (mass server access) is suspicious, RAG context confirms: 1. The user `admin_user` is a privileged administrator with a history of this activity. 2. The activity matches a documented, recurring monthly patching script (`monthly_patch.ps1`). 3. The behavior is consistent with authorized administrative tasks, not a ransomware infection.”
- **Outcome:** The analyst confidently closes the alert as a false positive in seconds, saving significant investigation time and allowing them to focus on real threats.

4.5 Discussion

The case studies presented demonstrate the dual value of the SecAINW platform. The first case study illustrates its ability to detect novel, low-and-slow threats by using RAG to identify deviations from a learned baseline. The second case study highlights its power in reducing alert fatigue by correctly contextualizing benign, privileged activity that would otherwise trigger false positives. This ability to provide not just a verdict, but a detailed, evidence-backed explanation, is what bridges the gap between raw detection and actionable

intelligence. The following sections will address the practicalities of reproducing this work and the limitations identified during the evaluation.

4.6 Reproducibility

The SecAINW platform was built using publicly available frameworks and an open-source dataset to ensure that the results can be reproduced. The architecture, detailed in the Experimental Setup (Section 4.1), relies on Angular, Spring Boot, Spring AI, and the LANL dataset. The complete, commented source code for both the `secainw-frontend` and `secainw-backend` modules is available in the project’s repository; please see **Appendix A**. A live demonstration of the platform, executing the evaluation test set sequences detailed in **Appendix B** in real-time and showcasing the analyst feedback loop, will be presented as part of this research project’s defense.

4.7 Discussion and Limitations

The results of this evaluation confirm the core hypothesis of this research: that a framework combining LLMs, RAG, and XAI can successfully address the critical challenges of static thresholds and poor interpretability in modern IDS. The findings in **Table 1** and **Table 2** show a clear and measurable improvement in both detection accuracy and analyst-rated usefulness over baseline methods.

However, this study also identified several significant limitations that must be addressed for future real-world deployment.

- **Performance and Latency:** The most critical limitation is the high end-to-end latency. As shown in **Table 3**, all tested models, including the fastest (Anthropic at ~29s), far exceeded the <15-second target for practical analyst workflows. This latency makes the current implementation unsuitable for real-time, inline detection and must be the primary focus of future optimization. Potential solutions include model quantization, parallelizing API calls, or fine-tuning smaller, task-specific models.

- **Data Dependency:** The system’s intelligence is entirely dependent on the quality and comprehensiveness of the data in the RAG knowledge base. A vector store populated with incomplete or “noisy” data will provide poor context, leading to incorrect verdicts and explanations.
- **Scalability:** This evaluation was conducted on a small, static test set ($n = 10$). The computational cost, data ingestion throughput, and query performance of the Oracle 23ai vector database and LLM APIs at the scale of a true enterprise network (processing millions or billions of events per day) are unknown.
- **Generalizability:** The framework was trained and tested exclusively on the LANL dataset. Its effectiveness against different network architectures, data sources (e.g., cloud logs, endpoint logs), and novel threat vectors not present in the LANL data has not yet been validated.

A notable limitation of this evaluation is the size of the test set ($n = 10$ events), following up from the scalability consideration above. While this sample size limits the statistical significance regarding the system’s performance on high-volume traffic, the test set was rigorously curated to function as a representative behavioral cross-section. As detailed in **Appendix B**, the dataset intentionally maximizes protocol diversity by encompassing HTTPS, SSH, SMB, and PowerShell execution to evaluate the model’s reasoning capabilities across distinct layers of the OSI model. Consequently, the evaluation validates the framework’s logical efficacy in detecting complex, multi-stage intrusions (such as the lateral movement in Event 8) and context switching (as seen in the temporal gap before Event 10), rather than its volumetric throughput.

5 FUTURE WORK

Building upon the current SecAINW framework and the insights gained from the evaluation (Section 4), this section outlines potential directions for future research and development. These directions aim to address the identified limitations and further enhance the platform’s capabilities and practical applicability in real-world Cybersecurity scenarios.

5.1 Addressing Current Limitations

The evaluation highlighted several key areas requiring improvement, which form the immediate priorities for future work:

- **Performance Optimization (Latency Reduction):** The most significant limitation identified was the high end-to-end latency, exceeding the practical <15-second target for analyst workflows (Section 4.7). Future work must prioritize drastically reducing this latency. Potential approaches include:
 - *Model Optimization:* Exploring techniques like model quantization, knowledge distillation to smaller, faster models, or fine-tuning task-specific LLMs optimized for speed.
 - *Parallel Processing:* Re-architecting the back-end to process requests to multiple LLMs concurrently rather than sequentially.
 - *Caching Strategies:* Implementing intelligent caching mechanisms for RAG retrievals and potentially even for common LLM responses or explanation patterns.
 - *Hardware Acceleration:* Investigating the use of specialized hardware (GPUs/TPUs) for LLM inference, potentially via optimized cloud instances or on-premise solutions.
- **Enhancing Data Quality and Scope:** The framework’s effectiveness is contingent on the RAG knowledge base (Section 4.7). Future efforts should focus on:
 - *Data Cleaning and Enrichment:* Developing more sophisticated ETL processes to improve the quality, normalization, and semantic coherence of ingested data chunks.

- *Expanding Data Sources*: Integrating additional data types beyond the LANL dataset, such as Cloud logs (AWS CloudTrail, Azure Activity Logs), endpoint detection and response (EDR) logs, and threat intelligence feeds, to broaden the system’s contextual understanding.
- **Scalability Testing and Improvement**: The current evaluation used a small test set (Section 4.7). Rigorous scalability testing is needed to assess performance under enterprise-level load. This involves:
 - *Volumetric Load Testing*: Simulating high volumes of concurrent anomaly detection requests.
 - *Database Optimization*: Evaluating and optimizing the query performance and indexing strategies of the Oracle 23ai vector store at scale.
 - *Distributed Architecture*: Exploring microservices or other distributed architectures for the back-end to handle increased load.
- **Generalizability Validation**: Testing the framework against diverse datasets and network environments is crucial (Section 4.7). This includes evaluating its performance on different types of network traffic, architectures, and against emerging threat vectors not represented in the initial LANL dataset.

5.2 Additional Future Directions

Beyond addressing current limitations, several other avenues could significantly enhance SecAINW’s utility:

- **Advanced XAI Techniques**: Integrating more sophisticated XAI methods, such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) adapted for LLMs, could provide deeper insights into model decisions beyond Chain-of-Thought reasoning. Visualizing attention maps or feature importance could also aid analyst understanding.

- **Automated Feedback and Reinforcement Learning:** Developing mechanisms for semi-automated feedback generation (e.g., correlating verdicts with subsequent incident response outcomes) could accelerate model improvement via RLHF/GRPO with less reliance on direct analyst input.
- **Proactive Threat Hunting Support:** Extending the RAG querying capabilities to allow analysts to perform natural language queries against the historical data store for proactive threat hunting, beyond just analyzing specific events.
- **Integration with SOAR Platforms:** Developing APIs and integrations to connect SecAINW with Security Orchestration, Automation, and Response (SOAR) platforms, enabling automated response actions based on high-confidence, explained detections.
- **Comparative LLM Analysis Framework:** Building functionality to more easily swap, compare, and ensemble different LLMs within the platform, allowing organizations to leverage the best model for specific types of security data or threats.
- **User Interface Enhancements:** Further developing the Angular front-end to include more advanced visualizations, trend analysis, and customizable dashboards based on analyst feedback.

Pursuing these future work directions will strengthen SecAINW's capabilities, address its current shortcomings, and enhance its potential as a valuable tool for security analysts in combating increasingly sophisticated network-related threats.

6 ETHICAL CONSIDERATIONS

The development and deployment of AI Systems in critical domains like Cybersecurity necessitate careful consideration of ethical implications. While SecAINW aims to enhance network security through advanced anomaly detection and explanation, several ethical factors must be addressed to ensure responsible implementation and use.

Potential for Bias: AI models, including LLMs, are susceptible to inheriting biases present in their training data. The SecAINW framework relies initially on the LANL Unified Host and Network Data Set. While valuable, this dataset represents a specific network environment during a particular time frame. Biases inherent in this data (e.g., skewed representation of certain activities, user groups, or network segments) could lead the LLM to exhibit unfair or inaccurate detection patterns. For instance, legitimate activities performed predominantly by a minority user group might be disproportionately flagged as anomalous if underrepresented in the training data used for the RAG knowledge base or the LLM's pre-training. Continuous monitoring for biased performance and exploring techniques for bias mitigation in data ingestion and model fine-tuning are crucial future steps.

Data Privacy: The SecAINW system processes potentially sensitive network flow and host event data to detect anomalies. In a real-world deployment, this data could contain information related to individual user activities, communication patterns, or system configurations. It is imperative that robust data privacy measures are implemented. This includes ensuring compliance with relevant privacy regulations (e.g., GDPR, CCPA), employing data anonymization or pseudonymization techniques where feasible during the ETL process, and implementing strict access controls for both the raw data and the RAG vector store. The system should be designed to minimize the collection and retention of Personally Identifiable Information (PII).

Transparency and Accountability: A core objective of SecAINW is to improve interpretability through XAI. This is not merely a functional requirement but an ethical one.

Security analysts must be able to understand *why* the system flags an event to trust its judgment and make informed decisions. The use of techniques like Chain-of-Thought reasoning aims to provide this transparency. However, the completeness and faithfulness of these explanations must be continually evaluated. Ensuring that the system's reasoning is auditable is vital for accountability, especially in cases where an AI-driven decision leads to significant consequences (e.g., blocking critical traffic based on a false positive).

Over-Reliance and Deskilling: As AI tools become more capable, there is a risk that Security Analysts may become overly reliant on them, potentially leading to a decline in their own critical analysis skills. If SecAINW consistently provides accurate verdicts and explanations, analysts might implicitly trust its output without sufficient scrutiny. This could lead to missed detections if the AI encounters a novel threat outside its learned patterns or fails due to unforeseen circumstances (e.g., data poisoning). Training programs and operational procedures should emphasize that AI tools like SecAINW are decision *support* systems, augmenting rather than replacing human expertise and critical judgment.

Dual Use: While SecAINW is designed for Defensive Cybersecurity purposes (anomaly detection), the underlying AI technologies (LLMs, RAG) could potentially be adapted for malicious uses if not properly secured. Ensuring the security of the model, the data, and the platform itself is paramount to prevent misuse.

Addressing these ethical considerations proactively throughout the design, development, deployment, and maintenance lifecycle of SecAINW is essential for building a trustworthy and responsible AI-driven security solution. Continuous ethical reflection and stakeholder engagement will be necessary as the technology evolves.

7 CONCLUSION

This research project introduced SecAINW, a novel data-driven security AI framework designed to enhance Intrusion Detection Systems (IDS) by integrating Large Language Models (LLMs), Retrieval Augmented Generation (RAG), and Explainable AI (XAI). The primary motivation was to address the persistent challenges in traditional anomaly detection, namely the difficulty in establishing effective detection thresholds and the lack of interpretability in AI-driven security alerts. By leveraging RAG to provide historical context from the LANL dataset and utilizing LLMs (DeepSeek, OpenAI, Anthropic) for nuanced analysis, the framework demonstrated a significant improvement in detection accuracy compared to a non-RAG baseline, as shown in the evaluation results (Section 4.2.1).

A key contribution of this work is the successful integration of XAI techniques, particularly evident in the high analyst ratings for explanation quality (Section 4.2.2). The framework's ability to provide clear, human-readable justifications for its verdicts, exemplified in the case studies (Section 4.4), directly addresses the need for transparency and trust in AI security tools. This moves beyond simple detection towards actionable intelligence, empowering security analysts to understand the reasoning behind alerts and respond more effectively. The case studies specifically highlighted SecAINW's potential both in identifying subtle, novel threats and in reducing false positives by correctly contextualizing benign administrative activities.

Despite the promising results in detection efficacy and interpretability, the evaluation also identified critical limitations, most notably the high processing latency of the current LLM integrations, which hinders real-time applicability (Section 4.7). Furthermore, considerations regarding potential data bias, scalability, generalizability, and ethical deployment remain pertinent (Section 6, Section 4.7).

Future work will focus on optimizing performance, expanding data source integration, enhancing XAI methods, and conducting rigorous scalability testing (Section 5). While

challenges remain, the SecAINW framework represents a significant step towards more adaptive, context-aware, and interpretable AI-Driven Cybersecurity. Its novel approach holds considerable potential for improving security posture by enabling faster, more informed incident response and potentially uncovering previously undetected intrusions through the synergistic power of RAG, LLMs, and XAI. The project repository and planned demonstration ensure the reproducibility of this research (Section 4.6). Ultimately, this work contributes to the evolving landscape of AI in Cybersecurity, offering a promising direction for future IDS development.

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Appendix A

CODE REPOSITORIES FOR REPRODUCIBILITY

To support the reproducibility of this research, the complete source code for the SecAINW platform is available in two public GitHub repositories. The frontend repository contains the Angular application, and the backend repository contains the Spring Boot application with all API logic and LLM integrations. Table 4 lists the repository details.

Table 4
GitHub Repositories for SecAINW

Repository	Description & URL
SecAINW Frontend	Contains the Angular 20+ source code for the analyst dashboard, visualization components, and feedback interface. URL: https://github.com/christophervi/secainw-frontend
SecAINW Backend	Contains the Spring Boot 3.4+ source code, including the Spring AI integration for LLMs, RAG implementation, CVE API client, and all REST API endpoints. URL: https://github.com/christophervi/secainw-backend

Appendix B

EVALUATION EVENT TEST SET

The evaluation results presented in Section 4 were based on a curated test set of 10 network events. Table 5 lists the identifiers, titles, and raw event data for this test set.

Table 5
List of 10 Network Events Used in Evaluation

Event ID	Event Title	Raw Event Data
INT-WEB-001	Normal Internal Web Access	HTTPS connection established to internal site.
EXT-WEB-002	Normal External Web Access	TLSv1.3 handshake successful.
SCAN-TCP-001	External Port Scan (Suspicious)	TCP SYN packet received, no response.
SCAN-TCP-002	External Port Scan - Different Port (Suspicious)	TCP SYN packet received, no response.
APACHE-CONN-001	External Connection to Web Server - Apache	GET /?query=<script>alert(1)</script> HTTP/1.1
HOST-PS-001	Unusual PowerShell Execution on Web Server	powershell -nop -c "IEX(New-Object Net.WebClient).DownloadString(...)"
OUTBOUND-C2-001	Outbound Connection from Web Server to Attacker	TCP connection established, high byte count outgoing.
LATERAL-SSH-001	Internal Lateral Movement Attempt	SSH authentication attempt failed for user 'root'.
EXCH-CONN-001	Microsoft Exchange Server Interaction	POST /ews/exchange.asmx HTTP/1.1 User-Agent: python-requests/2.25.1
INT-FS-003	Normal Internal File Server Access	SMB Read Request, File: \\fileserver\share\document.docx

Appendix C

FULL SAMPLE LLM EXPLANATIONS AND ANALYST RATINGS

To support the aggregated metrics in **Table 2**, this appendix provides a complete representative sample set of XAI explanations generated by the three models for all 10 evaluation events, along with the corresponding analyst quality ratings, as presented in **Tables 6 through 15**.

Table 6
Comparative Explanations for Event 1 (Normal Internal Web Access)

Model	Analyst Rating (Explain Quality)	Generated Explanation (XAI)
DeepSeek	4 / 5	<i>"Normal web traffic detected. Source IP 192.168.1.150 is accessing internal server 192.168.1.20. Protocol is HTTPS (Port 443). No malicious indicators found."</i>
OpenAI	5 / 5	<i>"Verdict: Normal. The source workstation (192.168.1.150) is establishing a secure HTTPS connection to the internal web server (192.168.1.20). RAG context confirms this is a routine connection between authorized internal assets."</i>
Anthropic	5 / 5	<i>"Verdict: Normal (High Confidence). **Reasoning:** 1. **Context:** Internal workstation accessing internal web server. 2. **Action:** Standard HTTPS connection established using 'chrome.exe'. 3. **RAG:** No anomalies detected in history for this source-destination pair."</i>

Table 7
Comparative Explanations for Event 2 (Normal External Web Access)

Model	Analyst Rating (Explain Quality)	Generated Explanation (XAI)
DeepSeek	4 / 5	<i>"Normal behavior. Browsing external site (104.18.32.101). Firefox process used. TLSv1.3 handshake successful, indicating encrypted traffic."</i>
OpenAI	5 / 5	<i>"Normal event. User on 192.168.1.150 connected to external IP 104.18.32.101 via Firefox. The successful TLSv1.3 handshake indicates secure, standard browsing activity to a legitimate external domain."</i>
Anthropic	5 / 5	<i>"Verdict: Normal. **Reasoning:** 1. **Source:** Internal workstation (192.168.1.150). 2. **Destination:** A known benign external web IP (Content Delivery Network). 3. **Process:** 'firefox.exe' is an authorized browser. 4. **Assessment:** Activity aligns with standard employee internet usage baseline."</i>

Table 8
Comparative Explanations for Event 3 (External Port Scan)

Model	Analyst Rating (Explain Quality)	Generated Explanation (XAI)
DeepSeek	3 / 5	<i>"Suspicious.</i> <i>TCP SYN packet received with no response.</i> <i>External IP 118.25.6.178 is scanning port 22 (SSH)."</i>
OpenAI	4 / 5	<i>"Suspicious activity detected.</i> <i>External IP 118.25.6.178 sent a TCP SYN packet to port 22 (SSH) on your web server but received no response.</i> <i>This resembles a reconnaissance port scan attempting to identify open services."</i>
Anthropic	4 / 5	<i>"Verdict: Suspicious.</i> <i>**Analysis:**</i> <i>1. **Actor:** Unknown External Host 118.25.6.178.</i> <i>2. **Action:** Probing port 22 (SSH) with a SYN packet.</i> <i>3. **Outcome:** The lack of response (no SYN-ACK) suggests the port is closed or filtered, but the intent is likely malicious reconnaissance."</i>

Table 9
Comparative Explanations for Event 4 (External Port Scan - Different Port)

Model	Analyst Rating (Explain Quality)	Generated Explanation (XAI)
DeepSeek	3 / 5	<i>"Suspicious.</i> <i>Same IP (118.25.6.178) is now scanning port 3389 (RDP).</i> <i>Suspicious scanning behavior confirmed."</i>
OpenAI	4 / 5	<i>"Suspicious.</i> <i>The same external IP (118.25.6.178) observed in Event 3 is now probing port 3389 (RDP).</i> <i>Sequential probing of different high-value ports (SSH, then RDP) confirms active scanning behavior."</i>
Anthropic	5 / 5	<i>"Verdict: Suspicious.</i> <i>**Correlation:**</i> <i>This event follows Event 3 immediately (within 5 seconds).</i> <i>**Pattern:** The attacker is shifting targets from SSH to RDP.</i> <i>**Conclusion:** This sequential scanning confirms hostile intent to find any open entry vectors on the public server."</i>

Table 10
Comparative Explanations for Event 5 (Apache XSS Attempt)

Model	Analyst Rating (Explain Quality)	Generated Explanation (XAI)
DeepSeek	4 / 5	<i>"Attack detected.</i> <i>XSS payload found in URL parameters.</i> <i>The string contains <script>alert (1) </script>.</i> <i>Source IP matches previous scanner."</i>
OpenAI	5 / 5	<i>"Anomalous/Attack.</i> <i>Malicious payload detected in HTTP GET request.</i> <i>The string <script>alert (1) </script> is a classic Cross-Site Scripting (XSS) probe attempting to execute code on the Apache server.</i> <i>This is a clear escalation from the previous port scanning."</i>
Anthropic	4 / 5	<i>"Verdict: Anomalous.</i> <i>**Threat:** Web Attack (Reflected XSS).</i> <i>**Evidence:** The URL contains a script tag payload.</i> <i>**Context:** This targets the public web server's HTTP service.</i> <i>RAG confirms this source IP has been scanning ports 22 and 3389 previously."</i>

Table 11
Comparative Explanations for Event 6 (Unusual PowerShell Execution)

Model	Analyst Rating (Explain Quality)	Generated Explanation (XAI)
DeepSeek	3 / 5	<i>"Malicious PowerShell. Downloading script from external IP 118.25.6.178. Hidden window flag used."</i>
OpenAI	4 / 5	<i>"Critical Anomaly. PowerShell execution with '-nop' flags detected on the Web Server. It attempts to download 'payload.ps1' from the known attacker IP 118.25.6.178 via IEX. This is a signature of fileless malware downloaders and requires immediate response."</i>
Anthropic	5 / 5	<i>"Verdict: Anomalous (High Severity). **1. Technique:** PowerShell downloading remote content (IEX/DownloadString). **2. IOC:** The remote IP 118.25.6.178 matches the previous scanner/attacker. **3. Assessment:** Confirmed Dropper/Downloader execution attempting to install malware or a backdoor."</i>

Table 12
Comparative Explanations for Event 7 (Outbound Connection to Attacker)

Model	Analyst Rating (Explain Quality)	Generated Explanation (XAI)
DeepSeek	4 / 5	<i>"Anomaly.</i> <i>Outbound connection initiated by Web Server to attacker IP.</i> <i>High data transfer on port 4444.</i> <i>Likely a reverse shell."</i>
OpenAI	5 / 5	<i>"Anomalous.</i> <i>Web server initiated a TCP connection to 118.25.6.178 on port 4444.</i> <i>This is a non-standard port often used for Reverse Shells (C2).</i> <i>The high outgoing byte count indicates significant data exfiltration is occurring."</i>
Anthropic	4 / 5	<i>"Verdict: Anomalous.</i> <i>**Stage:** Command and Control (C2).</i> <i>**Evidence:** The web server is connecting back to the attacker's IP on port 4444 immediately after the PowerShell execution.</i> <i>**Impact:** The high outbound traffic volume suggests active data exfiltration or a stable shell session."</i>

Table 13
Comparative Explanations for Event 8 (Internal Lateral Movement)

Model	Analyst Rating (Explain Quality)	Generated Explanation (XAI)
DeepSeek	2 / 5	<i>"Anomaly. SSH attempt failed. User 'root'. IP 203.0.113.50. IP 192.168.1.10. This is bad."</i>
OpenAI	4 / 5	<i>"Event [LATERAL-SSH-001] is anomalous. A failed root login via SSH from your web server (203.0.113.50) to an internal database (192.168.1.10) was detected. Web servers should not be initiating SSH to databases as 'root'. This indicates a lateral movement attempt."</i>
Anthropic	5 / 5	<i>"Verdict: Anomalous (High Confidence). Reasoning: 1. **Actor:** The SSH connection originated from a public-facing web server (203.0.113.50), which is a high-value target for compromise. 2. **Action:** The connection attempted to log in as 'root', which violates security best practices and is a common tactic for attackers. 3. **Target:** The destination (192.168.1.10) is an internal database server. 4. **Conclusion:** This event is a classic indicator of attempted lateral movement."</i>

Table 14
Comparative Explanations for Event 9 (Exchange Server Interaction)

Model	Analyst Rating (Explain Quality)	Generated Explanation (XAI)
DeepSeek	4 / 5	<i>"Suspicious Anomaly.</i> <i>1. Source: Compromised Web Server.</i> <i>2. Target: Exchange Server.</i> <i>3. Indicator: User-Agent 'python-requests' is highly abnormal for EWS traffic.</i> <i>4. Conclusion: Likely automated exploitation script running internally."</i>
OpenAI	4 / 5	<i>"Suspicious.</i> <i>Internal web server is accessing the Exchange server's EWS endpoint.</i> <i>The User-Agent 'python-requests' is abnormal for standard Exchange OWA/EWS traffic, which usually comes from browsers or Outlook.</i> <i>This suggests an automated tool or script is being used."</i>
Anthropic	4 / 5	<i>"Verdict: Suspicious.</i> <i>**Anomaly:** Web Server talking to Exchange Server.</i> <i>**Tool:** Python script usage detected via User-Agent.</i> <i>**Context:** RAG retrieval shows no prior history of the Web Server communicating with Exchange, indicating a deviation from established server-to-server communication patterns."</i>

Table 15
Comparative Explanations for Event 10 (Normal Internal File Access)

Model	Analyst Rating (Explain Quality)	Generated Explanation (XAI)
DeepSeek	5 / 5	<i>"Normal.</i> <i>User accessing file share via SMB protocol.</i> <i>Source: Workstation 192.168.1.151.</i> <i>Target: File Server 192.168.1.40.</i> <i>No threats detected; matches allowed business logic."</i>
OpenAI	4 / 5	<i>"Normal.</i> <i>Host 192.168.1.151 is reading 'document.docx' from the internal file server.</i> <i>This matches standard workflow for this user and subnet.</i> <i>No signatures of ransomware (e.g., mass writes or encryption extensions) were observed."</i>
Anthropic	4 / 5	<i>"Verdict: Normal.</i> <i>**Behavior:** Standard SMB file read operation.</i> <i>**User:** Authorized workstation.</i> <i>**Context:** Activity aligns with baseline business hours and file access patterns retrieved from the RAG knowledge base."</i>

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